

Simulating Logistics Systems

Lesson 3/15

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Outline of this lecture

Activity Cycle Diagrams: Examples

Python Practice

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Activity Cycle Diagrams: Examples

Python Practice

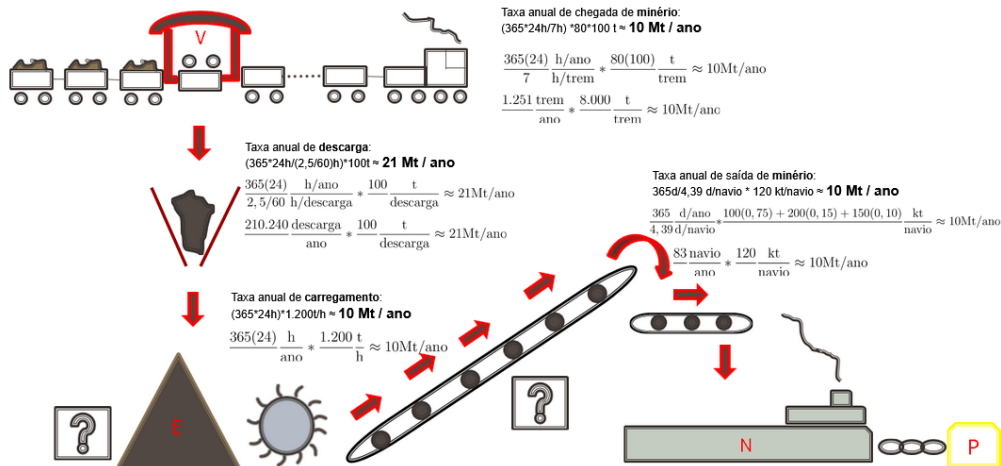
ACD: Example 4 [Pinto, 2016]

Port. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ An ore shipping port has only one pier and operates as follows: **Ships arrive** at the port at intervals that follow an exponential distribution with an average of **4.39 days**. The capacity of the ships varies as follows: **75%** are *100,000 tons*, **15%** are *200,000 tons*, and **10%** are *150,000 tons*.
- ▶ Upon arriving at the port, **ships** find the pier vacant and, provided there is ore in stock, begin to be **loaded** at a rate of **1,200 tons/hour**. If the ore stock runs out before the ship is completely loaded, the ship will wait at the pier for the arrival of ore, meaning it will only leave the pier when it is completely loaded. **The ore** arrives at the port by rail, with each **train** consisting of **80 wagons** with a capacity of **100 tons each**. Trains **arrive** at the port on average every 7 hours, following a normal distribution with a standard deviation of **1.07 hours**. Upon arrival at the port, the trains have their wagons unloaded one by one by a single wagon turner. The **unloading time for each car** follows a normal distribution with an average of **2.5 minutes** and a **standard deviation of 0.3 minutes**. At the end of each car's unloading, the port's ore stock obviously increases by 100 tons.
- ▶ Construct the ACD for this system.

ACD: Example 4 [Pinto, 2016]

Port. (calculating **average annual rates**) Dynamic presentation (Pt-Br) in Google Sheets

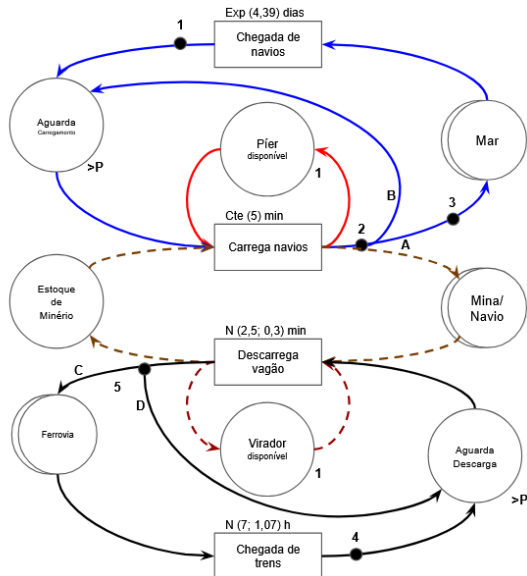
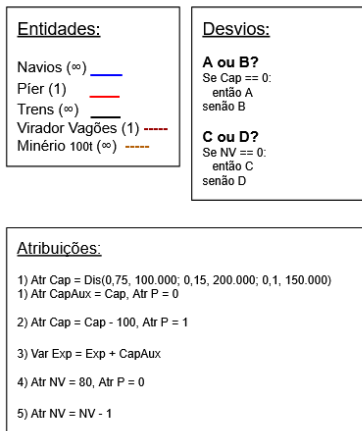


Port. (Considerations for modeling)

- ▶ The ore will be discretized into **100t** (1 railcar load);
- ▶ The ore entity = **100 t** → Each railcar carries *1 ore entity*;
- ▶ The **ship** loading activity will be discretized into *loads of 100t*;
- ▶ The ship loading rate is 1,200 t/h, so the load will take $(100 \text{ t} / 1,200 \text{ t/h}) * 60 \text{ min} = 5 \text{ min}$. That is, *the ship loading time: 5 min / ore entity*.

ACD: Example 4 [Pinto, 2016]

Port. Dynamic presentation (Pt-Br) in Google Sheets

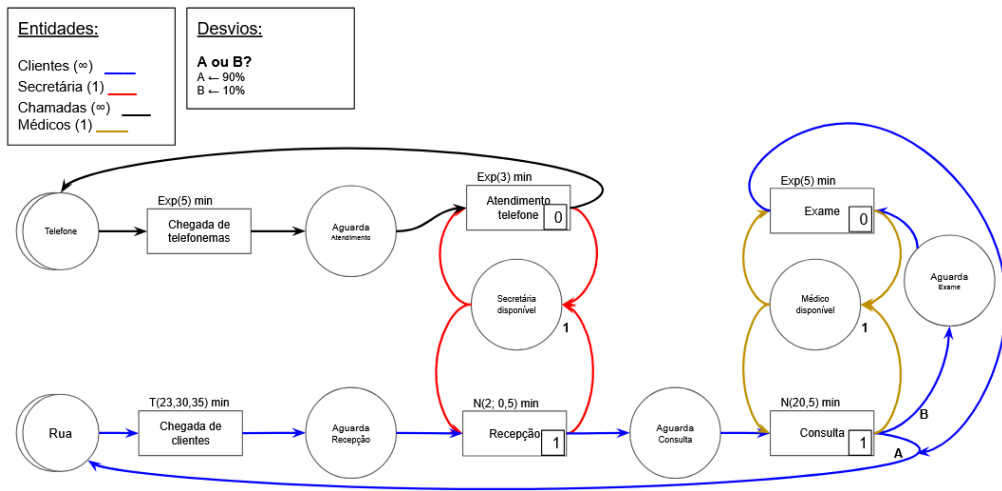


ACD: Example 5 [Pinto, 2016]

Doctor's office. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ A doctor's office operates as follows (all time values in minutes): patients arrive according to a triangular distribution with minimum, mode and maximum of 23, 30, and 35. Upon arrival, they are assisted by a secretary who fills out an electronic form with patient information. The time taken for this service follows a normal distribution with a mean of 2 and a standard deviation of 0.5.
- ▶ Once the form has been filled out, the patient waits for their appointment with the doctor, its duration follows a normal distribution with a mean of 20 and a standard deviation of 5. After the appointment, 10% of patients undergo some examination at the office, while the rest leave. The exam is performed immediately after the consultation by the doctor himself, with an exponentially distributed duration with a mean of 5. After that, patients leave the office.
- ▶ Besides attending to patients and filling out forms, the secretary also answers the phone, with calls arriving at intervals in an exponential distribution with a mean of 5. The telephone conversation is exponentially distributed with a mean of 3. Telephone service, when the secretary is available, is a priority. When she is attending to a patient, she finishes the consultation before answering the phone.

Doctor's office. Dynamic presentation (Pt-Br) in Google Sheets



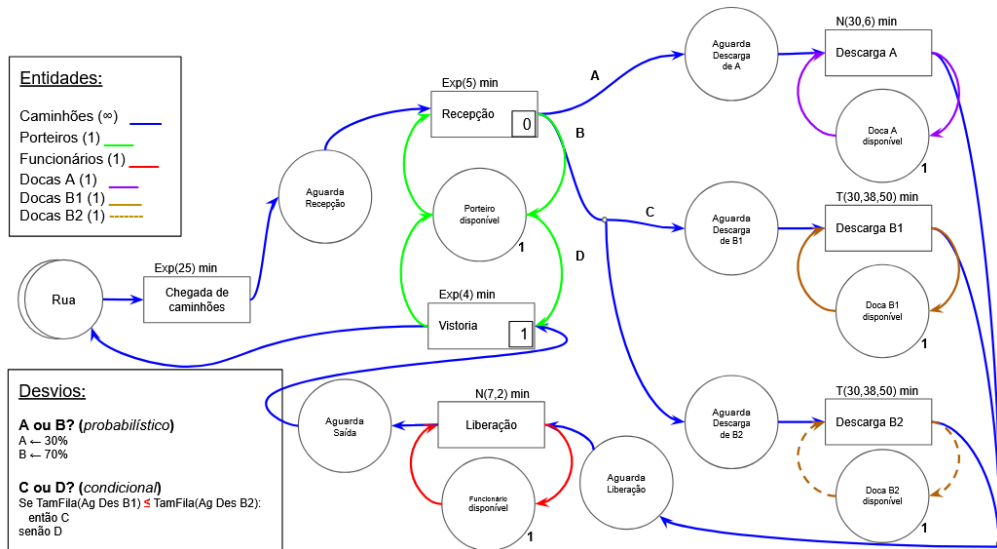
ACD: Example 6 [Pinto, 2016]

Transportation company. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ A company uses raw materials MPA and MPB, which arrive by truck at exp(25) minutes. 30% of trucks carry MPA; the rest carry MPB. At the company, trucks are checked by a gatekeeper, who directs them to an unloading dock, taking exp(5) minutes. The unloading time for trucks carrying MPA (Dock A) is normally distributed with a mean of 30 minutes and a standard deviation of 6 minutes; for MPB there is 2 docks (Docks B1 and B2), it is triangular with a mode of 38, a minimum of 30, and a maximum of 50 minutes. The trucks are directed to the dock with the shortest queue. After unloading, trucks proceed to another sector of the company to deliver invoices and unloading receipts. Here, they are attended by an employee, who fills out a vehicle release form. The time spent by the employee performing this service is normally distributed with a mean of 7 minutes and a standard deviation of 2 minutes. After receiving the release form, trucks proceed to the company's gate, where the same employee performs a safety inspection and then releases them. The time spent on this inspection is exponentially distributed with a mean of 4 minutes. The gate employee prioritizes the arrival of trucks over the exit inspection.

ACD: Example 6 [Pinto, 2016]

Transportation company. Dynamic presentation (Pt-Br) in Google Sheets



Transportation company. Dynamic presentation (Pt-Br) in Google Sheets

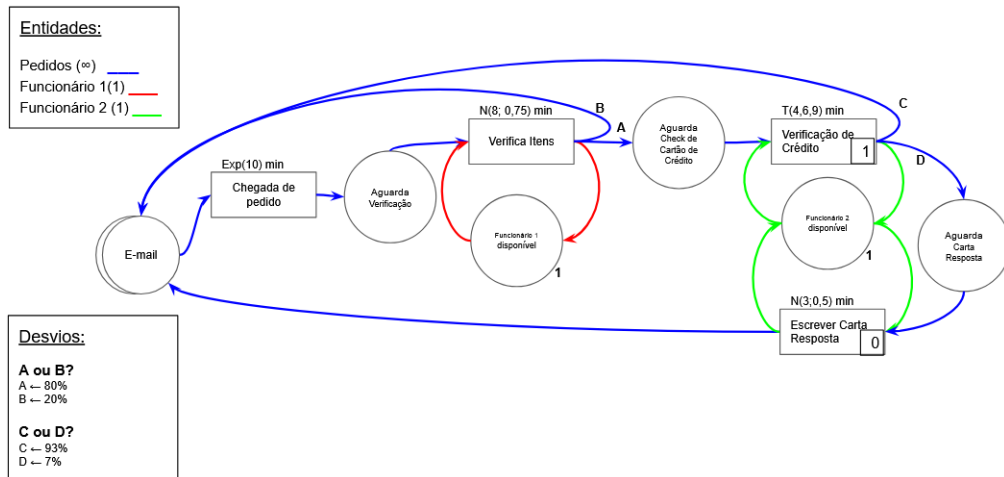
- ▶ **Exercise:** Analyze the $\leq$ sign present in deviations C and D (exercise 6) and answer which dock would be more overloaded and why? If the sign were only $<$, what could be expected?
- ▶ Redo Example 6 considering that, when trucks are directed to docks B1 and B2, they will enter a **single queue** and be unloaded at the dock that empties first.

ACD: Example 7 [Pinto, 2016]

E-commerce. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ In an e-commerce company, orders are emailed to the sales department, where an employee checks if all items are in stock. If any items are missing, the order is sent to the production department. If all items are available, the order is sent to another employee, who contacts the credit card company to verify that the purchase can be charged to the customer's credit card. If there is a problem with the card, the order is refused and the employee writes and sends a message to the customer.
- ▶ Orders arrive at 10-minute intervals. Stock verification time follows a normal distribution with an average of 8 minutes and a standard deviation of 0.75 minutes. Credit verification follows a triangular distribution with a minimum of 4, a mode of 6, and a maximum of 9 minutes. The time to write and send the message to the customer when the order is declined by the card issuer follows a normal distribution with a mean of 3 minutes and a standard deviation of 0.5 minutes.
- ▶ It is known that historically, 20% of orders contain missing items and that 7% of card transactions are declined by the administrator.

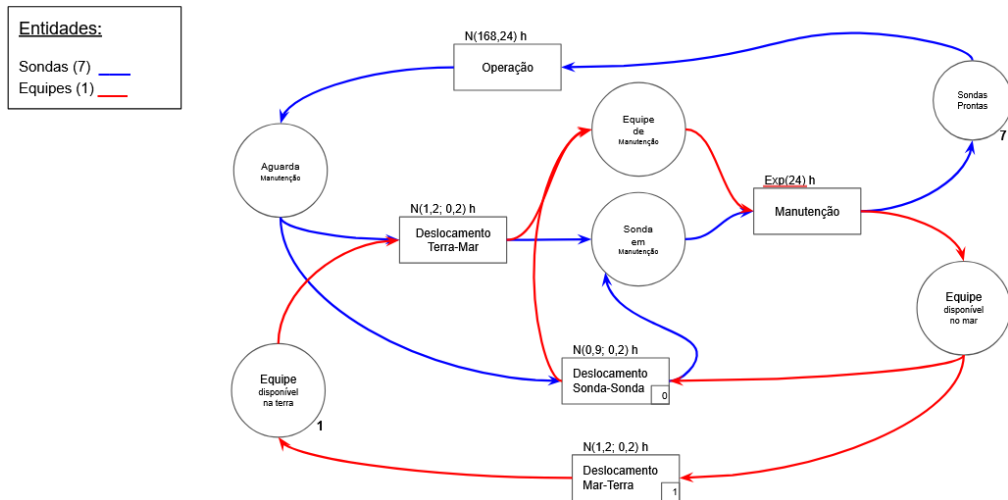
E-commerce. Dynamic presentation (Pt-Br) in Google Sheets



Petroleum exploration drilling rigs. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ A company operates seven oil drilling rigs in an offshore oil field. The rigs **operate continuously**, with their operation interrupted only for corrective maintenance. The **interval between failures** is described by a normal distribution with a **mean of 168 hours and a standard deviation of 24 hours**. **Maintenance** is performed by a **single team** and its duration is exponentially distributed, with a **mean of 24 hours**.
- ▶ At the start of the operation, the team is based onshore. Whenever a rig breaks down, **the team travels** to the rig's location, where it remains until maintenance is complete. At the end of maintenance, if there are no more broken rigs, **the team returns to base**. **If** there is another broken probe, the team travels directly to the probe that has been waiting the longest for maintenance. **Travel times between rigs** are described by a normal distribution, with a **mean of 0.9 h and a standard deviation of 0.2 h**. **Travel times between rigs and the ground base** also follow a normal distribution, with a **mean of 1.2 h and a standard deviation of 0.2 h**. That said, it is necessary to make the ACD representative of the system.

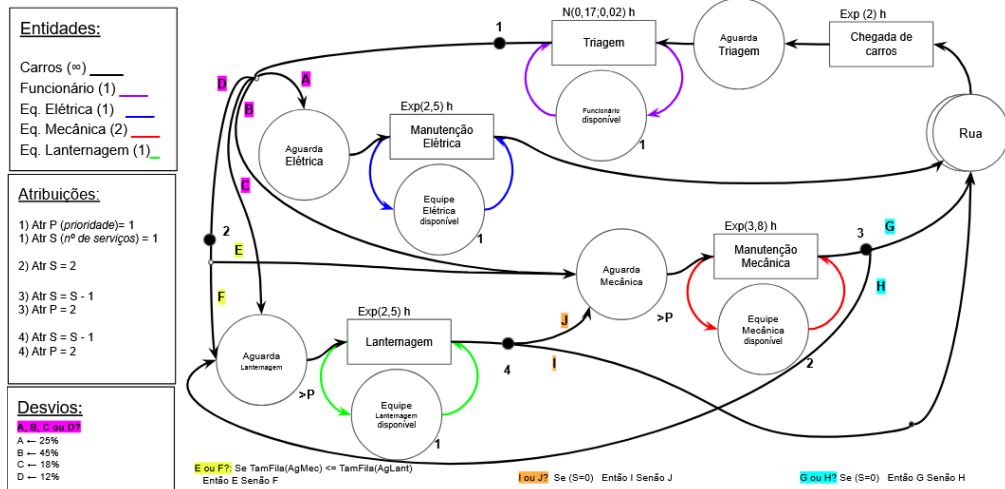
Petroleum exploration drills. Dynamic presentation (Pt-Br) in Google Sheets



Car repair shop. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ An auto repair shop performs mechanical, electrical, and bodywork services. It has 4 teams: 2 for *mechanical*, 1 for *electrical*, and 1 for *bodywork*. Cars undergo an initial screening by a single employee and are sent for service. 45% require mechanical, 25% electrical, 18% bodywork, and 12% both mechanical and bodywork. The latter (12%) are assisted at the shortest waiting line service, then forwarded to the other service, with priority over vehicles in the waiting line.
- ▶ Car arrivals follow an exponential distribution with an average of 2 hours. Triage time follows a normal distribution with an average of 0.17 hours and a standard deviation of 0.02 hours. Mechanical maintenance time follows an exponential distribution with an average of 3.8 hours. Electrical maintenance time follows an exponential distribution with an average of 2.5 hours. Bodywork time follows an exponential distribution with an average of 5 hours. Construct the system's ACD.

Car repair shop. Dynamic presentation (Pt-Br) in Google Sheets



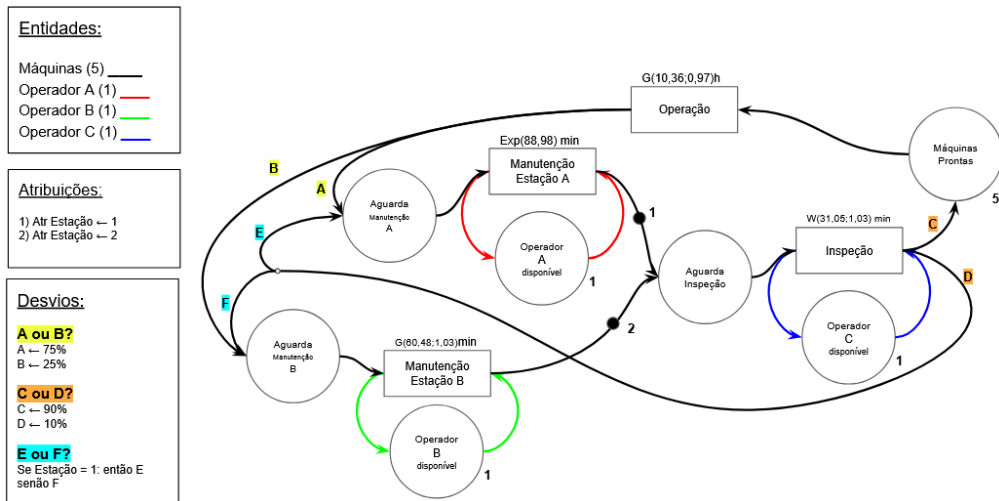
ACD: Example 10 [Pinto, 2016]

Maintenance of internal and external machinery. Dynamic presentation (Pt-Br) in Google Sheets

- ▶ A company has a maintenance workshop with five machines used for operations within its industrial area. Inside the workshop, there are two repair stations, station A and station B. At each station, there is 1 operator available to perform the work. The probability of a machine needing repairs at station A is 75% and at station B is 25%. After being repaired, a machine goes for a final inspection, where there is a single operator. After inspection, 90% of the machines are released for operation and 10% return for further maintenance. This further maintenance always takes place at the same station where the machine was initially repaired. In addition to maintaining the company's machines, this workshop is also studying the possibility of providing services to third parties, i.e. maintenance on machines from other companies. External machines would always be repaired at station B and, after repair, would also be inspected by the same operator and released (82% of cases) or retained for further maintenance (18% of cases). In this case, the new maintenance would always take place at station B. Times follow the distributions: Inspection: Weibull (31.05, 1.03) min; Repair Station A: Exp (88.98) min; Repair Station B: Gamma (60.48, 1.03) min; Interval between failures: Gamma (10.36, 0.97)h; Interval between arrivals of external machines: Gamma (7.97, 0.96)h. Construct one ACD with internal machines only, and another ACD with internal and external machines.

ACD: Example 10.1 [Pinto, 2016]

Maintenance of internal and external machinery. Dynamic presentation (Pt-Br) in Google Sheets



Outline of this lecture

Activity Cycle Diagrams: Examples

Python Practice

Python Practice

- ▶ Download (or clone) *the course Github repository*;
- ▶ Create a virtual environment: `py -m venv venv`
- ▶ Activate the virtual environment: `./venv/Scripts/activate.ps1` (**venv**)
- ▶ Install requirements: `pip install -r requirements.txt`
- ▶ Install ipython: `pip install ipython`
- ▶ Open 00-Supplementary-Material-Python-Course
- ▶ Run Ipython: `> ipython`
- ▶ Open a file to study: e.g. 01-Sintaxe.py
- ▶ Select each line (or more) and SHIF + ENTER

References



Pinto, L. R. (2016).

Simulação por eventos discretos.

Notas de Aula., 1.